

Results

01 · ANCOVA

group means adjusted for a covariate

Table 1. Adjusted (estimated marginal) means

Group	Adj. M	SE	95% CI
control	31.42	1.11	[29.20, 33.65]
drug_a	35.02	1.11	[32.80, 37.25]
drug_b	37.69	1.10	[35.48, 39.90]

Table 2. ANCOVA

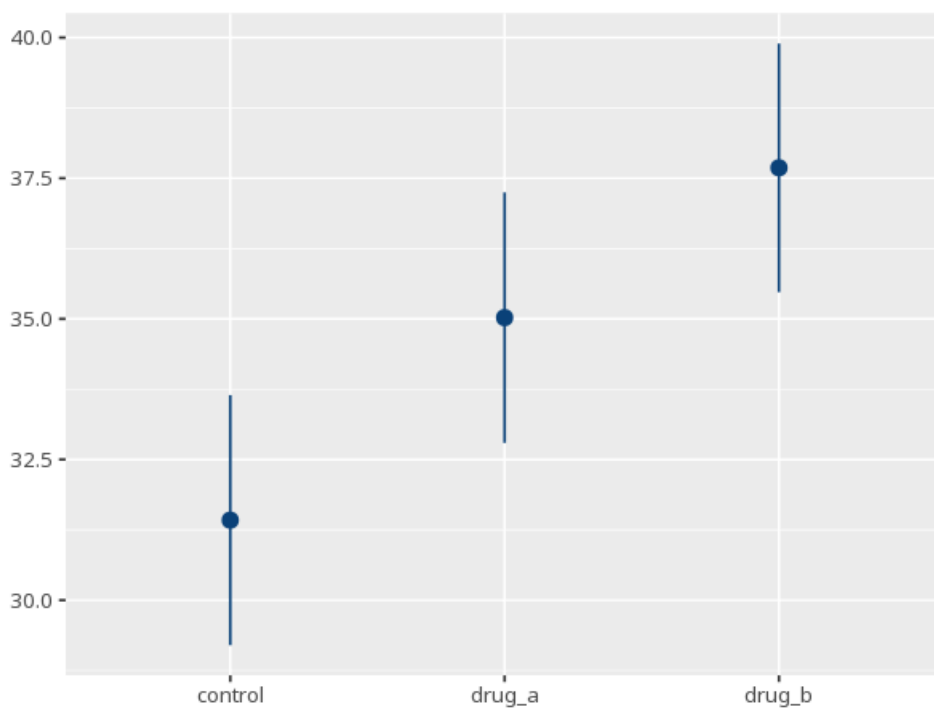
Source	SS	df	MS	F	p	partial η^2 [95% CI]
baseline	1757.40	1	1757.40	72.23	<.001	0.56 [0.42, 1.00]
group	392.60	2	196.30	8.07	<.001	0.22 [0.07, 1.00]

assumption checks: homogeneity of regression slopes (factor×covariate interaction), Levene's, & residual normality (Shapiro-Wilk); post-hoc on adjusted means. (slopes $p(\text{baseline} \times \text{group})=.875$ · Levene $F=0.02$, $p=.978$ · Shapiro $W=0.98$, $p=.475$) — the regression slopes look homogeneous across groups — equal variances look reasonable — residual normality looks reasonable

Table 3. Post-hoc comparisons (adjusted means)

Pair	M_{diff} (adj.)	SE	P_{adj}	95% CI
control - drug_a	−3.60	1.58	.068	[−7.41, 0.21]
control - drug_b	−6.26	1.56	<.001	[−10.03, −2.50]
drug_a - drug_b	−2.66	1.57	.214	[−6.44, 1.11]

Figure. Adjusted means



Understanding the numbers

Source. Names which term the ANCOVA row tests - a covariate or the group factor. Here, the group factor's row is group.

SS. The variability attributable to that term, after the covariate is accounted for. Here, group has SS = 392.60.

df. The degrees of freedom for that term. Here, group has df = 2.

MS. That term's SS divided by its df. Here, group has MS = 196.30.

F. The ratio of that term's mean square to the residual mean square, after removing the covariate's influence. Here, F for group = 8.07.

p. The probability of an F this large for the group factor if, after adjusting for the covariate, the groups truly had equal means. Here, $p < .001$ for group - below the $\alpha = 0.05$ threshold.

partial η^2 . The share of variance the group factor explains after removing the covariate's influence. Here, partial η^2 for group = .22 [.07, 1.00].

Adj. M. The group's estimated mean after statistically removing the covariate's influence. Here, each of the 3 groups has its own adjusted mean in the adjusted-means table above.

SE. The standard error of that adjusted mean, or of a pairwise adjusted-mean difference. Here, each adjusted mean and each pairwise contrast above carries its own SE.

95% CI. The range likely to contain the true adjusted mean, or true adjusted mean difference, for that row. Here, each adjusted mean and each pairwise contrast above has its own 95% CI shown in the tables.

M diff (adj.). The difference between two groups' adjusted means in the post-hoc table. Here, each post-hoc row above reports its own adjusted mean difference for a specific pair.

p (adj). The pairwise p-value for that adjusted-mean comparison, corrected for the number of comparisons made. Here, each pairwise contrast above carries its own multiple-comparison-adjusted p.

How to read this test

Compares group means after statistically removing the influence of a numeric covariate. First confirm the factor×covariate interaction is non-significant (homogeneity of slopes) — if it is significant, the adjusted-means interpretation is not valid and a different model is needed. Then read the Factor row's F/p for the adjusted group effect, and the adjusted means for the covariate-controlled group values. Your significance threshold (α) is 0.05.

APA template: Controlling for the covariate, an ANCOVA gave $F(2,56)=8.07$, $p < .001$, partial $\eta^2=.22$ [0.07, 1.00].

Statistical basis: ANCOVA (group means adjusted for a covariate) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); Type III sums of squares (car::Anova) (Fox J, Weisberg S (2019). An R Companion to Applied Regression, Third edition. Sage, Thousand Oaks CA.); Estimated marginal (covariate-adjusted) means (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.); Partial η^2 effect-size benchmarks (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.). Full references in the exported CITATIONS.txt.

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